

AN EXACT TWO-CHANNEL TRANSFORM AND A RECIPROCAL-SOURCE OBSTRUCTION FOR THE ERDŐS–STRAUS EQUATION

THE CLANKERS

ABSTRACT. On every negative-character source in the square-source ES system, exterior and middle hits require opposite original divisor characters. A common p -coloured pair source has exact fibres of size $2\tau_{\text{div}}(h)$, whose Möbius transform recovers the full joint count. The character forcing new source factors is annihilated in that count. We give a finite local obstruction to immediate reciprocal-source repair, and an original non-square exchange that crosses the channels elsewhere at the same prime. No universal ES nonvanishing result is asserted.

Let $p \equiv 1 \pmod{4}$ be prime, $p/4 < a < p/2$, $R = 4a - p$, and $u \mid a^2$. The original labelled gates are

$$E : R \mid 4u + 1, \quad M : R \mid 4u + p.$$

With $d = \gcd(a, u)$ and $(h, r, s) = (d^2/u, u/d, a/d)$ their ordered returns are

$$E : (a, h s \kappa, p h r \kappa), \quad \kappa = (pr + s)/R; \quad M : (a, p h s \lambda, p h r \lambda), \quad \lambda = (r + s)/R. \quad (1)$$

Their ordered inverses are $a^2/(Ry - pa)$ and $pa^2/(Ry - pa)$ respectively. These are inherited Type I/II coordinates [1]. Every coordinate is retained.

Proposition 1 (Channel characters). *For every original divisor, $(u/R) = (u/p)$. An E hit requires $(u/p) = -1$; an M hit requires $(u/p) = -(a/p)$. Thus on a negative source $(a/p) = -1$, unchanged- u and rational-square-ratio transports cannot cross E and M .*

Proof. For odd $\ell \mid a$, reciprocity and $R \equiv -p \pmod{\ell}$ give $(\ell/R) = (-1/\ell)(-p/\ell) = (\ell/p)$. The case 2 follows from $R \equiv -p \pmod{8}$ when $2 \mid a$. Multiply over the original exponents. The two gate residues are $-1/4$ and $-p/4$, and $(p/R) = (a/p)$. $\square$

In particular, for $a = (p + \sigma_q q t^2)/4$ with q an external nonresidue prime and a permitted hard-prime square source, the two required signs are opposite. This is a restriction on successful returns, not a sufficiency test.

For $\gcd(pN, R) = 1$, $p \nmid N$, retain the single ordered pair source

$$\mathcal{P}_R(N) = \{(b, c) : b, c \mid pN, R \mid b + c\}, \quad T_R(N) = |\mathcal{P}_R(N)|.$$

Let $F_R(N)$ count ordered coprime pairs $rs \mid N$ satisfying $R \mid r + s$ (label M) or $R \mid pr + s$ (label E), as a disjoint tagged union.

Theorem 2 (Exact fibres and joint count). *At a primitive pair put $h = N/(rs)$. Its complete fibre is*

$$\begin{aligned} M : (p^\varepsilon dr, p^\varepsilon ds), \quad d \mid h, \quad \varepsilon = 0, 1; \\ E : (pdr, ds) \text{ or } (ds, pdr), \quad d \mid h. \end{aligned}$$

It has size $2\tau_{\text{div}}(h)$. Consequently

$$F_R(N) = \frac{1}{2} \sum_{d \mid N} \mu_M(d) T_R(N/d). \quad (2)$$

At the actual shell $N = a$, $F_R(a) = |E_a| + |M_a|$, so the complete first-half count is

$$\mathcal{C}_p = \frac{1}{2} \sum_{p/4 < a < p/2} \sum_{d \mid a} \mu_M(d) T_{4a-p}(a/d). \quad (3)$$

Proof. Divide (b, c) by its gcd. If neither coprime part contains p , remove the common p -bit and recover the first displayed fibre. If exactly one contains p , remove it, retain its side, and recover the second. The lcm condition gives $d \mid h$. These operations are inverse. Thus $T_R(N)/2 = \sum_{d \mid N} F_R(N/d)$; Möbius inversion follows from $\sum_{d \mid n} \mu_M(d) = \mathbf{1}_{n=1}$. At $N = a$, the inverse $u = hr^2$ gives exactly the original states. Auxiliary $N = a/d$ are divisor-pair sources, not shells satisfying $4N = p + R$. $\square$

For existence, the exact fibres also give $2F_R(a) \leq T_R(a) \leq 2\tau_{\text{div}}(a)F_R(a)$. Hence (3) is positive exactly when $\sum_{p/4 < a < p/2} T_{4a-p}(a) > 0$. One can force an original pair first and normalize afterward; termwise positivity of the Möbius summands is unnecessary.

The two p -colours give the matrix

$$\begin{pmatrix} B_0 & B_1 \\ B_1 & B_0 \end{pmatrix}, \quad B_0 = \#\{b, c \mid N : R \mid b + c\}, \quad B_1 = \#\{b, c \mid N : R \mid b + pc\}.$$

It need not be positive semidefinite. At $p = 193, R = 7, a = 50$, its entries are $B_0 = 4, B_1 = 6$; twenty ordered pairs normalize to four original states, not ten.

Character orthogonality and the local Fejér difference give

$$F_R(N) = \frac{1}{2\varphi(R)} \sum_{\chi \bmod R} \chi(-1) |1 + \chi(p)|^2 \prod_{q^e \parallel N} \sum_{j=-e}^e \chi(q)^j. \quad (4)$$

Indeed $T_R(N)$ has the same expression with the last product replaced by $|\sum_{d \mid N} \chi(d)|^2$; subtracting the previous prime exponent gives $|1 + z + \dots + z^e|^2 - |1 + z + \dots + z^{e-1}|^2 = \sum_{j=-e}^e z^j$. At every negative square source the transported Jacobi character satisfies $\chi_R(p) = -1$, so its entire term is killed by $|1 + \chi_R(p)|^2$. Thus the character forcing factors does not itself force the combined target. The other signed terms remain an arithmetic obligation.

Proposition 3 (Reciprocal source and exact inverse). *For $D \equiv 3 \pmod{4}$, let $B = (pD + 1)/4$. A divisor $w \mid B^2$ with $D \mid w + B$ normalizes as $B = hrs$, $w = hr^2$, $\gcd(r, s) = 1$ and $r + s = D\lambda$. After recording the swap making $r < s$, it returns*

$$(x, y, z) = (hr\lambda, hs\lambda, pB), \quad p/4 < x < p/2.$$

It is the E state $(h, r, s_{\text{original}}, \kappa) = (h, r, \lambda, s)$. The inverse is $D = (4u + 1)/R$, $B = hr\kappa$, with the two source divisors $w = hr^2$ and $h\kappa^2$.

Proof. Use $4hrs = pD + 1$ and $r + s = D\lambda$. The original residual satisfies $(4x - p)s = pr + \lambda > 0$. Moreover $(4hrs - 1)(s - r) - (r + s) = 2s(2hr(s - r) - 1) > 0$, proving the strict upper bound. The displayed identities solve back for D, B and both orientations. The symmetric master equation is inherited [2], not a new existence assertion. $\square$

Every first-half E state has $D \leq \lfloor (p^2 + 6p + 13)/12 \rfloor$, by $u \leq a^2$ and maximizing $(4a^2 + 1)/(4a - p)$ at residual 3.

Proposition 4 (Overlap supplies only the old endpoint). *For $p \equiv 1 \pmod{8}$, $0 < D < 3p$, put $A = (p + D)/4$, $B = (pD + 1)/4$. A nonresidue prime dividing both must be $\ell \equiv 3 \pmod{4}$ with $\ell \mid p + 1$. Conversely any such factor gives a common nonresidue overlap for some $D < p$.*

Proof. $\gcd(A, B) = \gcd(A, (D^2 - 1)/4)$. At a common prime $D = \pm 1$ and $p = -D$; reciprocity rules out the sign $p = 1$ and requires $\ell \equiv 3 \pmod{4}$ otherwise. For the converse take the least such ℓ . Since $(p + 1)/2 \equiv 1 \pmod{4}$, $\ell^2 \leq (p + 1)/2$; then $D = 2\ell + 1 < p$ gives the overlap. The original endpoint E return is $R = \ell$, $a = u = (p + \ell)/4$, $\kappa = (p + 1)/\ell$. $\square$

At the certified prime $p = 12889$, consider $q = 43, 61$ and all their valid square ports $D = \sigma_q q t^2 < 3p$: respectively $t = 1, 3, \dots, 29$ and $t = 1, 3, \dots, 13$. The complete finite certificate exhausts the 246 original divisors in the 22 direct sources (the inherited Turn 4 fixture [3]) and the 600 original divisors in all 22 reciprocal sources. All direct E/M and all reciprocal sum gates are empty. This is 44 boxes and 66 gate families, not a whole-graph or ES counterexample. At the canonical ports the reciprocal sources are $127 \cdot 1091$ and $2^3 \cdot 73709$, with no nonresidue factors. The two-cycle therefore remains closed under that canonical-only reciprocal augmentation. A reciprocal nonresidue factor is not sufficient either: $B_{387} = 37 \cdot 33703$ has two, but its gate is empty.

At the same p , the genuine crossing occurs at $R = 31$, $a = 3230$:

$$u_M = 25, \quad u_E = 85, \quad \frac{u_E}{u_M} = \frac{17}{5} \equiv p^{-1} \pmod{31}.$$

This ratio has character -1 and stays within the actual prime-exponent box. The returns are $(3230, 174852174, 1353345)$ and $(3230, 1346910, 456850605)$. The reciprocal source at $q = 61, t = 11$ has factorization $2 \cdot 19 \cdot 31 \cdot 37 \cdot 1637$ and supplies the actual nonresidue factor 31 leading to that occupied source. This is a path, not a universal path theorem.

The next target is a positive lower bound for the complete original sum (3). No such bound follows from (4)'s annihilated source character or from the formal reciprocal inverse. The standard-library certificate and separate implementation state their finite bounds and retain the original order, characters and fibres. No ES resolution, leastness, independent review, Lean build or historical priority is claimed.

REFERENCES

- [1] C. Elsholtz and T. Tao (2013), *Counting the number of solutions to the Erdős–Straus equation on unit fractions*, arXiv:1107.1010, §2.
- [2] C. Bryan (2026), *FAB-DUAL-DESCENT-SYSTEM.md*, CENTL revision 42508684c4386b575d7b52e763565345a76bcb9e, §§1–4.
- [3] The Clankers (2026), *Turn 4: mixed square-source finite reduction*, PR19 revision 4b6bb0f187fa44bb62fc8c51e996f05dfc90e2d7, §8.